

# **ABES Engineering College, Ghaziabad**

**Department of ASH**



Estd. 2000

## **Lab Manual**

**Year/ Sem.** : 1<sup>st</sup> year/ I sem.

**Subject Name** : Web Designing

**Subject Code** : 25VA151

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Approved by AICTE, New Delhi & Affiliated to Dr. APJ Abdul Kalam Technical University, UP,  
Lucknow

NBA Accredited Courses (CSE, IT, ECE, EN)

19<sup>th</sup> KM Stone, NH-24, Ghaziabad-201009, UP



# ABES Engineering College, Ghaziabad

## VISION

To take ABES Engineering College to such a level that, it is at par with the leading institutions of the world in providing leadership to the international education system and be amongst the top-rated institutions of the world by providing a transformative education to create leaders and innovators embedded in traditional Indian values.

## MISSION

- To create an ambiance for healthy teaching learning process.
- To nurture the students and infuse in them:
  - A passion to excel professionally
  - A spirit to be of utmost use to the industry, corporate sector and the society at large
  - An intense desire to take challenging responsibilities and leadership roles
  - A craving to be wholesome good human beings
- To develop an environment for creating new knowledge through research and by thriving to explore innovative ideas.

## QUALITY POLICY

To continuously thrive to provide a congenial and wholesome academic environment and a healthy culture for faculty, staff and students which would motivate teacher's full participation with passion and develop an intense desire in the students to acquire comprehensive education and hence become a useful and confident human resource for the industry and academia.

## VALUES

**Respect** – We treat people the way we want to be treated

**Integrity** – We believe in doing things right when no one is watching

**Accountability** – We take full responsibility of our actions and are accountable for them

**Transparency** – We communicate openly amongst ourselves and all our stakeholders to avoid any misconception.

**Excellence** – We endeavor to think out-of-the-box and deliver maximum value to all our Stakeholders.



# **ABES Engineering College, Ghaziabad**

## **Department of ASH**

### **VISION**

The department of ASH has a distinct vision to nurture and inculcate knowledge and understanding of engineering sciences amongst the budding technocrats and groom the students with intellectual and professional strength. ASH is an anvil to chisel future professionals with the skills of engineering, imbued with social, ethical and moral values, so that the pupils can expedite their efforts to build technologies from knowledge and be a part of the workforce, instrumental in the development of the nation's technological and economic progress.

### **MISSION**

M1: To play a pivotal role in laying the foundations of engineering education by building the professional strengths of the students in basic sciences and engineering disciplines and strengthening the foundation of aspiring engineers.

M2: To provide a unique learning environment for the students, preparing the students at par with the global technical workforce, building confidence, character and human values.

M3: To upgrade the quality of technical education and produce dynamic engineers and entrepreneurs for the country, with a sole aim to spot ABESEC amongst the best educational institutions of global standards



# ABES Engineering College, Ghaziabad

## Department of ASH

### **GENERAL RULES & INSTRUCTIONS OF LAB**

1. Students are advised to come to the laboratory at least 5 minutes before (to the starting time), those who come after 5 minutes will not be allowed into the lab.
2. Plan your task properly much before to the commencement, come prepared to the lab with the synopsis / program / experiment details.
3. Student should enter into the laboratory with:
  - a) Laboratory observation notes with all the details (Problem statement, Aim, Algorithm, Procedure, Program, Expected Output, etc.,) filled in for the lab session.
  - b) Laboratory Record updated up to the last session experiments and other utensils (if any) needed in the lab.
  - c) Proper Dress code and Identity card.

#### **Do's**

1. Switch off mobile phone in the lab.
2. Bring your lab record with you.
3. Always come to the lab in time.
4. Maintain silence and be attentive.
5. Keep individual records of observations.
6. Always turn off the computer before leaving the lab.
7. Return all the material issued to you before leaving the lab.
8. Read the instructions manual carefully before performing any experiment.

#### **Don'ts**

1. Eat or drink inside the lab.
2. Overcrowd near any equipment.
3. Make any mark on instruction manual.
4. Drag the chair/stool inside the lab.
5. Leave the lab till the class has concluded.
6. Take any material outside the lab with you.
7. Touch any machine/equipment/model without supervision.

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| Sr. No | Title of Experiment                                                                                                                                                                                  | KL     | CO  | Software Used                                                                                                                                 | Page No |
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| 1      | Create a Web Page for Course Curriculum: Use of Headings, Paragraphs, Bold, Italic, Underline, and Special Characters, Ordered List, Unordered List, Description List, Nested List, and Mixed Lists. | K2     | CO1 | <ul style="list-style-type: none"> <li>Text Editor (VS Code, Sublime Text, Notepad++)</li> <li>Web Browser (Chrome, Edge, Firefox)</li> </ul> |         |
| 2      | Create a Web Page for a News Website with Embedded YouTube Video, Hyperlinks, and Navigation Structure, and Insert Multimedia in a Webpage.                                                          | K2, K3 | CO1 | <ul style="list-style-type: none"> <li>Text Editor (VS Code, Sublime Text, Notepad++)</li> <li>Web Browser (Chrome, Edge, Firefox)</li> </ul> |         |
| 3      | Create Web page for College timetable: Use of tables with rowspan, colspan, headers, captions, and nested tables                                                                                     | K3     | CO2 | <ul style="list-style-type: none"> <li>Text Editor (VS Code, Sublime Text, Notepad++)</li> <li>Web Browser (Chrome, Edge, Firefox)</li> </ul> |         |
| 4      | Create Web page for Hotel room booking form: Use of input fields, labels, select, textarea, and buttons.                                                                                             | K3     | CO2 | <ul style="list-style-type: none"> <li>Text Editor (VS Code, Sublime Text, Notepad++)</li> <li>Web Browser (Chrome, Edge, Firefox)</li> </ul> |         |
| 5      | Create Web page for Student Registration Form with Email Validation: Apply form attributes (method, action, placeholder, required)                                                                   | K3     | CO2 | <ul style="list-style-type: none"> <li>Text Editor (VS Code, Sublime Text, Notepad++)</li> <li>Web Browser (Chrome, Edge, Firefox)</li> </ul> |         |
| 6      | Create Web page for E-commerce Product Page: Use of CSS for style text, colors, fonts, margin, border, padding, width, and height                                                                    | K3     | CO3 | <ul style="list-style-type: none"> <li>Text Editor (VS Code, Sublime Text, Notepad++)</li> <li>Web Browser (Chrome, Edge, Firefox)</li> </ul> |         |

|    |                                                                                                                       |               |            |                                                                                                                                                   |  |
|----|-----------------------------------------------------------------------------------------------------------------------|---------------|------------|---------------------------------------------------------------------------------------------------------------------------------------------------|--|
| 7  | Create Web page for Navigation Menu: .....Use of block, inline, inline-block, float, and clear                        | <b>K3, K4</b> | <b>CO4</b> | <ul style="list-style-type: none"> <li>• Text Editor (VS Code, Sublime Text, Notepad++)</li> <li>• Web Browser (Chrome, Edge, Firefox)</li> </ul> |  |
| 8  | Create Web page for Student Profile Card: design layouts using flexbox (justify-content, align-items, flex-direction) | <b>K4</b>     | <b>CO4</b> | <ul style="list-style-type: none"> <li>• Text Editor (VS Code, Sublime Text, Notepad++)</li> <li>• Web Browser (Chrome, Edge, Firefox)</li> </ul> |  |
| 9  | Create Web page for Responsive Product Grid: create mobile-friendly designs with breakpoints.                         | <b>K4</b>     | <b>CO4</b> | <ul style="list-style-type: none"> <li>• Text Editor (VS Code, Sublime Text, Notepad++)</li> <li>• Web Browser (Chrome, Edge, Firefox)</li> </ul> |  |
| 10 | Create Web page for E-commerce Style Navbar: create a responsive navbar with CSS and media queries                    | <b>K4</b>     | <b>CO4</b> | <ul style="list-style-type: none"> <li>• Text Editor (VS Code, Sublime Text, Notepad++)</li> <li>• Web Browser (Chrome, Edge, Firefox)</li> </ul> |  |

## Experiment-1

**Create a Web Page for Course Curriculum: Use of Headings, Paragraphs, Bold, Italic, Underline, and Special Characters, Ordered List, Unordered List, Description List, Nested List, and Mixed Lists.**

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### Objectives

1. To design a simple course curriculum webpage using HTML.
  2. To practice the use of basic text formatting tags (bold, italic, underline, special characters).
  3. To learn the implementation of ordered, unordered, description, nested, and mixed lists.
  4. To understand webpage content structuring using headings and paragraphs.
- 

### Tools Required

- Computer / Laptop with any OS
  - Text Editor (VS Code, Sublime Text, Notepad++)
  - Web Browser (Chrome, Edge, Firefox)
- 

### Theory

1. Headings (<h1>...<h6>): Used to display titles and subheadings in different sizes.
  2. Paragraphs (<p>): Defines blocks of text content.
  3. Formatting Tags:
    - <b> for bold text
    - <i> for *italic text*
    - <u> for <u>underlined text</u>
    - Special characters are displayed using HTML entities (e.g., &copy;, &lt;, &gt;).
  4. Lists:
    - Ordered List (<ol>): Numbered items.
    - Unordered List (<ul>): Bulleted items.
    - Description List (<dl>): Term and description pairs.
    - Nested List: A list inside another list.
    - Mixed List: Combination of ordered, unordered, and nested lists.
-

## Code Implementation

Students will attach the printout of the HTML code that produced the given output of Webpage.

## Output

|                                                                                                                                                                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Introduction</b><br>Welcome to the <b>Course Curriculum</b> . This page demonstrates the use of <i>HTML formatting</i> , <u>lists</u> , and special characters such as ©, <, >.                                                                                                                                                                |
| <b>Semester 1 Subjects (Ordered List)</b><br><ol style="list-style-type: none"><li>1. Mathematics I</li><li>2. Computer Fundamentals</li><li>3. Programming in C</li><li>4. Electronics Basics</li></ol>                                                                                                                                          |
| <b>Programming Languages (Unordered List)</b><br><ul style="list-style-type: none"><li>• C</li><li>• Java</li><li>• Python</li><li>• JavaScript</li></ul>                                                                                                                                                                                         |
| <b>Course Description (Description List)</b><br><b>C Programming</b><br>Learn basics of C language including loops, arrays, and functions.<br><b>Java</b><br>Introduction to Object-Oriented Programming concepts.                                                                                                                                |
| <b>Course Modules (Nested List)</b><br><ol style="list-style-type: none"><li>1. Module 1: Basics<ul style="list-style-type: none"><li>◦ Introduction</li><li>◦ History of Computers</li></ul></li><li>2. Module 2: Programming<ul style="list-style-type: none"><li>◦ Variables &amp; Data Types</li><li>◦ Control Statements</li></ul></li></ol> |
| <b>Project Work (Mixed List)</b><br><ul style="list-style-type: none"><li>• Web Development<ol style="list-style-type: none"><li>1. HTML</li><li>2. CSS</li><li>3. JavaScript</li></ol></li><li>• Database<ol style="list-style-type: none"><li>A. MySQL</li><li>B. MongoDB</li></ol></li></ul>                                                   |

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## Outcome

- Successfully designed a basic course curriculum webpage.
  - Implemented text formatting and different types of lists in HTML.
  - Learned structuring and organizing content on a webpage.
-

## Experiment 2

Create a Web Page for a News Website with Embedded YouTube Video, Hyperlinks, and Navigation Structure, and Insert Multimedia in a Webpage.

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### Objectives

1. To design a simple **news website homepage** using HTML and CSS.
  2. To learn how to **embed YouTube videos** into a webpage.
  3. To implement **hyperlinks and navigation menus** for easy browsing.
  4. To insert **multimedia (images, videos, audio)** in a webpage.
- 

### Tools Required

- Computer / Laptop with any OS
  - **Text Editor** (VS Code, Sublime Text, Notepad++)
  - **Web Browser** (Chrome, Edge, Firefox)
  - Internet connection (for YouTube embedding)
- 

### Theory

1. **Hyperlinks (<a> tag):**

Hyperlinks allow navigation between pages or websites. Example:

```
<a href="https://www.bbc.com">Visit BBC News</a>
```

2. **Navigation Structure:**

A <nav> element groups navigation links for better structure.

3. **Embedding YouTube Video:**

YouTube provides an <iframe> code for embedding videos into a webpage.

```
<iframe src="https://www.youtube.com/embed/VIDEO_ID"></iframe>
```

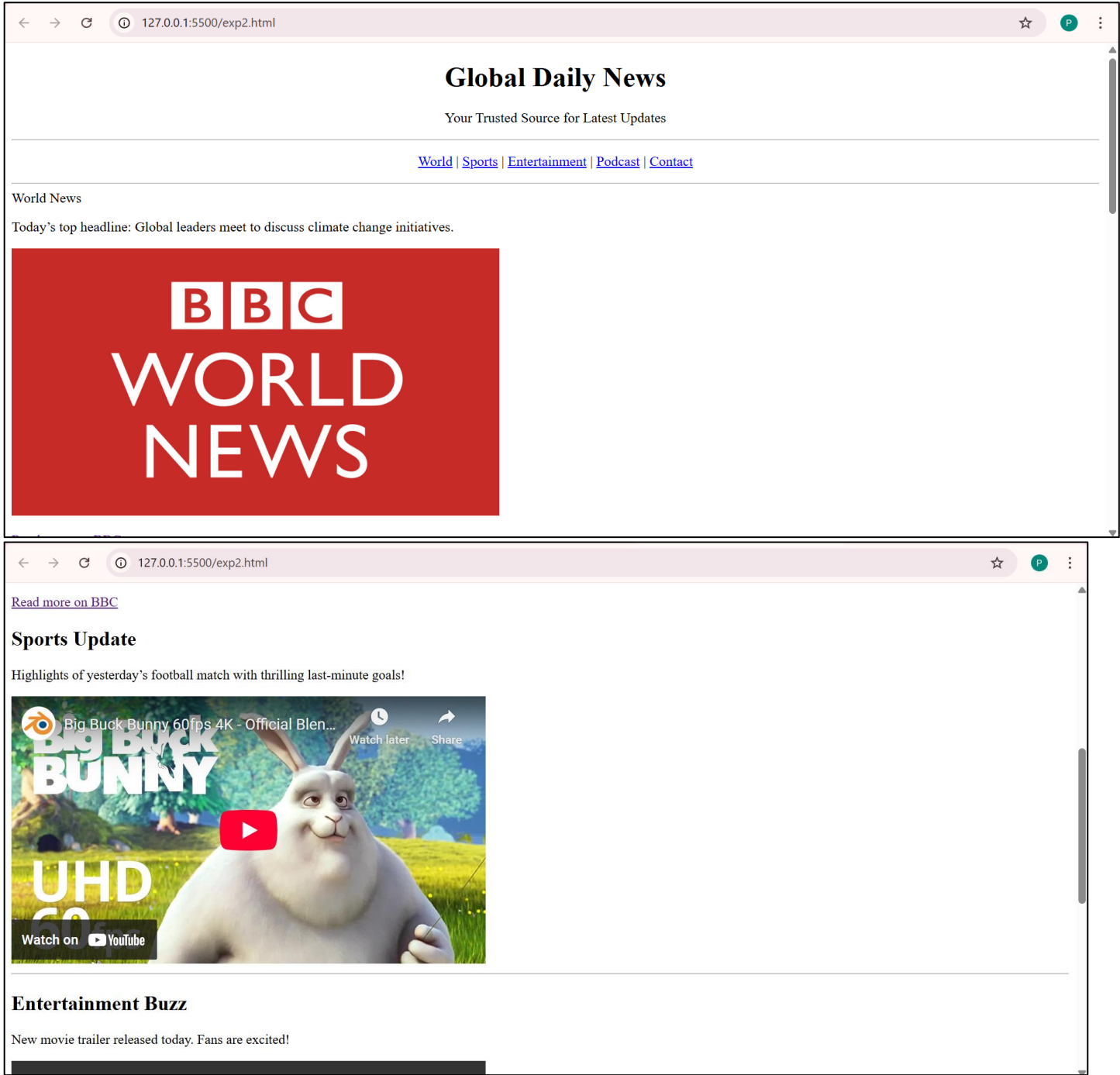
4. **Multimedia in Webpages:**

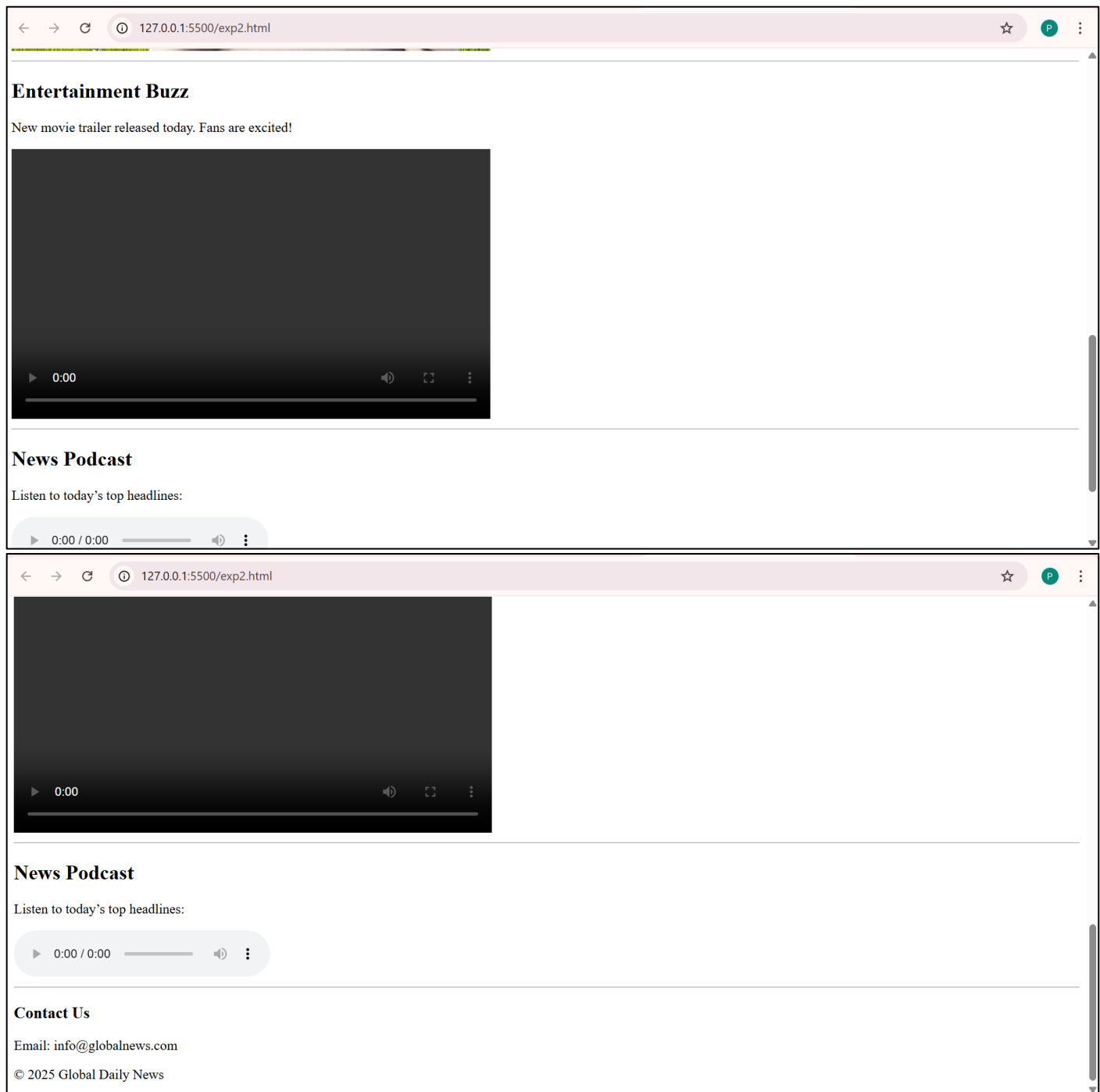
- **Image (<img>):** Display news images.
  - **Audio (<audio>):** Add podcasts or news audio clips.
  - **Video (<video>):** Insert local or hosted video files.
-

Code Implementation

Students will attach the printout of the HTML code that produced the given output of Webpage.

Output:





## Outcome

- Successfully designed a **basic news website** layout.
- Implemented **hyperlinks, navigation, multimedia (image, audio, video), and embedded YouTube video.**
- Learned practical **webpage structuring techniques** for media-rich sites.

## Experiment-3

### Create a Web Page for College Timetable: Use of Tables with Rowspan, Colspan, Headers, Captions, and Nested Tables.

#### Objectives

1. To design a college timetable webpage using HTML tables.
2. To apply rowspan and colspan attributes for merging cells.
3. To use table headers (<th>), captions (<caption>), and formatting for readability.
4. To implement nested tables within a timetable cell.

#### Tools Required

- Computer / Laptop with any OS
- Text Editor (Notepad, VS Code, Sublime Text)
- Web Browser (Chrome, Edge, Firefox)

#### Theory

1. Table (<table>): Arranges data in rows and columns.
2. Rowspan: Merges a cell vertically across rows.
3. <td rowspan="2">Lab</td>
4. Colspan: Merges a cell horizontally across columns.
5. <td colspan="3">Break</td>
6. Header (<th>): Displays header cells in bold and centered.
7. Caption (<caption>): Defines the title of the table.
8. Nested Table: A table placed inside another table cell.

#### Code Implementation

Students will attach the printout of the HTML code that produced webpage of the class timetable of your respective class.

#### Output:

| Day / Time  | 9:00 - 10:00      | 10:00 - 11:00 | 11:00 - 12:00 | 12:00 - 1:00 | 1:00 - 2:00       | 2:00 - 3:00                                                                                                                                | 3:00 - 4:00 |             |  |         |          |         |        |
|-------------|-------------------|---------------|---------------|--------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------|-------------|-------------|--|---------|----------|---------|--------|
| Monday      | Mathematics       | English       | Physics       | Break        | Computer Lab      |                                                                                                                                            | Sports      |             |  |         |          |         |        |
| Tuesday     |                   | C Programming |               | Break        | Electronics Lab   |                                                                                                                                            | Library     |             |  |         |          |         |        |
| Wednesday   | Mathematics       | English       |               | Break        | Computer Networks | <table><tr><th colspan="2">Lab Details</th></tr><tr><td>Batch A</td><td>DBMS Lab</td></tr><tr><td>Batch B</td><td>OS Lab</td></tr></table> |             | Lab Details |  | Batch A | DBMS Lab | Batch B | OS Lab |
| Lab Details |                   |               |               |              |                   |                                                                                                                                            |             |             |  |         |          |         |        |
| Batch A     | DBMS Lab          |               |               |              |                   |                                                                                                                                            |             |             |  |         |          |         |        |
| Batch B     | OS Lab            |               |               |              |                   |                                                                                                                                            |             |             |  |         |          |         |        |
| Thursday    | Workshop          |               | Physics       | Break        | Mathematics       | English                                                                                                                                    | Seminar     |             |  |         |          |         |        |
| Friday      | Computer Networks | English       | Physics       | Break        | Project Work      |                                                                                                                                            |             |             |  |         |          |         |        |

**Outcome:**

- Successfully created a college timetable webpage using HTML tables.
  - Implemented rowspan, colspan, headers, captions, and nested tables.
  - Learned to organize tabular data in a structured webpage format.
-

## Experiment-4

### Create Web page for Hotel room booking form: Use of input fields, labels, select, textarea, and buttons.

---

#### Objectives

1. To design a hotel room booking form using HTML.
  2. To understand the use of input fields for collecting user details.
  3. To apply labels for form accessibility.
  4. To implement dropdown menus (select) and text areas.
  5. To create functional buttons for form submission and reset.
- 

#### Tools Required

- Computer / Laptop with any OS
  - Text Editor (Notepad, VS Code, Sublime Text)
  - Web Browser (Chrome, Edge, Firefox)
- 

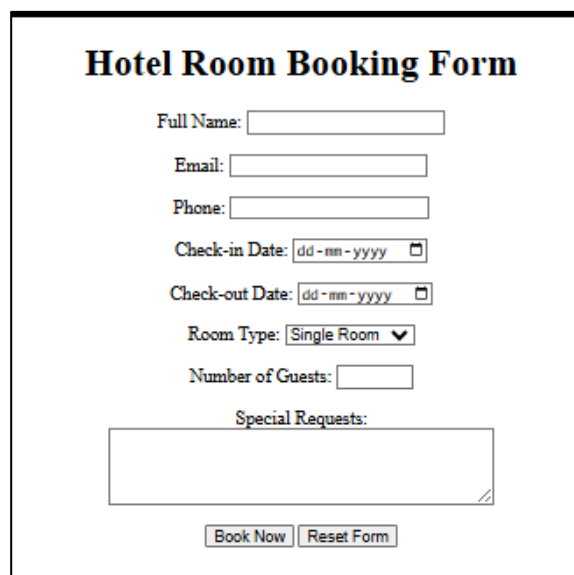
#### Theory

1. **Form (<form>):** Defines a section for user input.
  2. **Input Fields (<input>):** Used for text, numbers, emails, dates, etc.
  3. `<input type="text" name="username">`
  4. **Labels (<label>):** Connects text with form elements for better usability.
  5. `<label for="name">Name:</label>`
  6. **Select (<select>):** Provides a dropdown menu.
  7. `<select><option>Single</option><option>Double</option></select>`
  8. **Textarea (<textarea>):** Allows multi-line text input.
  9. **Buttons (<button> or <input type="submit">):** Used for form submission or reset.
- 

#### Code Implementation

Students will attach the printout of the HTML code that produced the given output of Webpage.

#### Output:



The screenshot displays a web form titled "Hotel Room Booking Form". It contains several input fields: "Full Name:", "Email:", "Phone:", "Check-in Date:" (with a date picker icon), "Check-out Date:" (with a date picker icon), "Room Type:" (a dropdown menu currently showing "Single Room"), and "Number of Guests:". Below these is a "Special Requests:" label followed by a large text area. At the bottom, there are two buttons: "Book Now" and "Reset Form".

**Outcome**

- Successfully created a hotel room booking form webpage using HTML.
  - Implemented input fields, labels, select, textarea, and buttons.
  - Learned practical form design techniques for web applications.
-

## Experiment 5

Create a Web Page for Student Registration Form with Email Validation by applying form attributes (method, action, placeholder, required).

---

### Objectives

1. To design a **student registration form** using HTML.
  2. To apply form attributes:
    - method → Defines how data is sent.
    - action → Defines where data is sent.
    - placeholder → Displays hint text.
    - required → Ensures mandatory fields are filled.
  3. To implement **basic email validation** using the type="email" attribute.
- 

### Tools Required

- Text Editor (Notepad, VS Code, Sublime Text)
  - Web Browser (Chrome, Edge, Firefox)
- 

### Theory

- **HTML Forms** (<form>): Used to collect user input.
  - **Form Attributes:**
    - method="post" → Sends form data securely.
    - action="submit.html" → Defines where the form data goes (dummy page here).
    - placeholder="Enter Name" → Shows hint text inside input box.
    - required → Makes the field compulsory.
  - **Email Validation:**
    - type="email" → Ensures user enters data in email format (e.g., abc@example.com).
- 

### Code Implementation

Students will attach the printout of the HTML code that produced the given output of Webpage.

# Output

127.0.0.1:5500/exp5.html

☆🔖📄⋮

# Student Registration Form

Full Name:

Email:

Password:

Gender: ☐ Male ☐ Female ☐ Other

Course: 

-- Select Course --

Contact No:

Address:

Enter your address

Register

Clear

© 2025 Student Registration Portal

# Outcome

- Successfully created a **Student Registration Form** using only HTML.
  - Applied **form attributes**: method, action, placeholder, required.
  - Implemented **email validation** using type="email".
  - Learned to design a **structured form** without CSS.
- 
-